LLM Finetuning

Pre-trained LLM model

- API (Groq, openAI)
- Downloading & loading in GPU (ollama)

→ Build GenAI Applications

Domain specific Tasks

Company Specific Tasks
org

Domain - Healthcare / Finance

→ Tool ← to provide the context

→ RAG context
↳ Data

Context is complex
Patterns are complex
} RAG Fails

Vector → Embedding Query
Data ← Embedding

Training model

Finetuning

Transfer learning

Transformers

W

↓

Update some of the weights

↓

Custom Data / smaller Dataset

↓

Pretrained model → few weight will be updated on smaller Dataset

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Fine-tuning → Open source

Full Fine-tuning

↓

LLM model

↓

Fine-tuning → weight → Dataset
} → Need larger infra

LORA → Low Rank Adaptation

transformer → Q, K V

Q - Query → W_Q
K - Key → W_K
V - value → W_V
O - Output → W_O
} W

LLM →



$W = W + \Delta W$

PEFT → Parameter Efficient Fine tuning

↳ LORA

↳ $W = W + \Delta W$ → Replica

VRAM

ΔW → We Dont train entire weight as it is

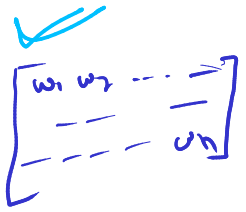
Rank

Rank = 8

Hyperparameter / intra

Finetune → Decompose matrix

Decompose



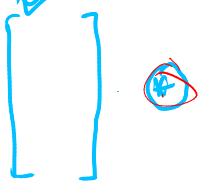
$\Delta W_{d \times d}$

d = 512

512 × 512 = 262144

rows columns 2D

3D



512 x 8

4096

[ ] ⇒ 8 × 512

4096

512 × 512

4096 + 4096 = 8192

Rank →

Rank ↑ → more parameters

Rank ↓ → less parameters

$W = W + \Delta W$

500 → 1 weight → 8 bit ⇒ 1 byte ⇒ 500 bytes

↓

VRAM

200 →

100 →

QLORA → Quantization

weight → Data type

500 weights

↳ weight → float 16 (High Precision) → 2 bytes

↳ Quantization

memory half → float 8 (Low Precision)

↳ 1

Float 8 | Int 8

float 16 | int-16

Float 32

[ ][ ][ ][ ][ ]

NFH → Normal float (4 bit)

4bit → in2

Original weights

w 0.23 → 2

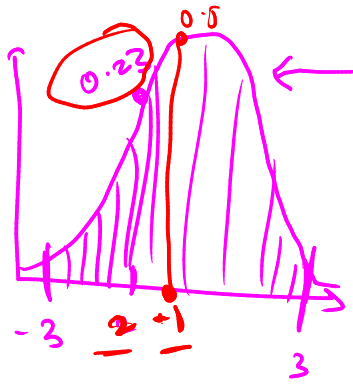
w = 0.5 → -1

[2, -1] ← Integer

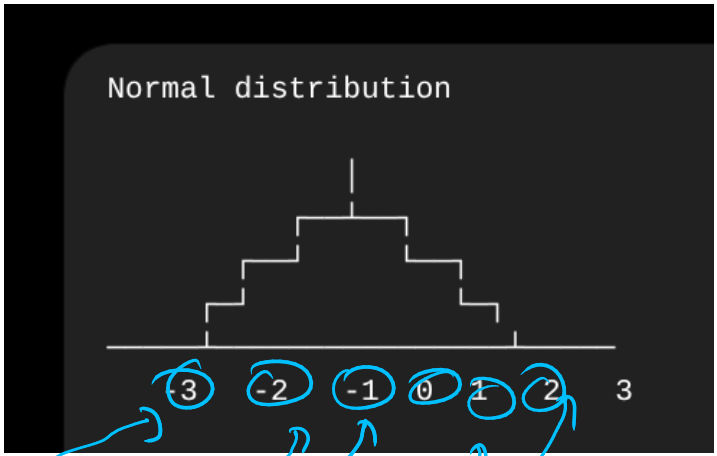
↓ De Quan

[0.23 0.5]

Normal Distribution weights







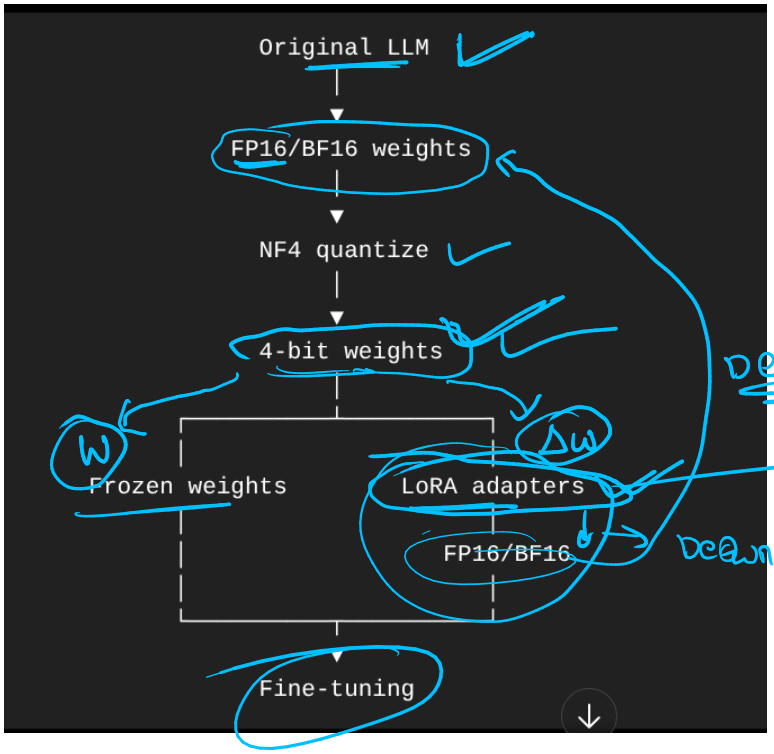
[-0.72, -0.31, 0.02, 0.45, 0.81, ...] ⇒ scaling

$w = \frac{w}{\text{Scale}}$

Scale = 0.81

[0, 3, 7, 12, 15, ...] ⇒ memory → Hbit

Scale



Dequantize

→ Decomposed Matrix

Dequant

SFT → Supervised Fine-tuning